Ski Route Planner

User Guide

1. Getting Started

The Ski Route Planner is a mobile-friendly web app for planning ski routes across multiple ski areas. It calculates step-by-step directions from any station to any other station, taking into account lift connections, piste difficulty, and current lift/piste closures.

Supported Ski Areas

The app currently supports two ski areas:

Ski Area	Sub-Areas	Coverage
Matterhorn Ski Paradise	Cervinia, Valtournenche, Zermatt	233 stations, 158 lifts, 240 pistes
Pontedilegno-Tonale	Ponte di Legno, Passo Tonale, Presena	57 stations, 62 lifts, 65 pistes

Opening the App

Open the app in your mobile or desktop browser. On first load you will see the Matterhorn Ski Paradise map with station markers. The app can be installed as a home-screen shortcut on iOS and Android for quick access.

2. Planning a Route

Adding Stops

There are two ways to add stops to your route:

- Tap a station marker on the map. The station is added as the next stop.
- Use the "Add stop" search box at the top. Type a station name to filter, then tap to select. Stations are grouped by sub-area for easy browsing.

You need at least two stops (a start and a destination) before a route is calculated. You can add as many intermediate stops as you like to create a multi-leg route.

Reordering & Removing Stops

Each stop in the list has controls to reorder or remove it:

- Tap the up/down arrows to move a stop earlier or later in the route.
- Tap the X button to remove a stop.

Setting Difficulty Preference

Below the waypoint list, choose your difficulty preference. This controls which pistes the router will use:

Setting	Behaviour
Blue	Only blue (easy) pistes. Safest option for beginners.
Blue/Red	Prefers blue pistes but will use red if no blue route exists.
Red	Uses blue and red pistes freely. The default setting.
Red/Black	Prefers red or easier, but will use black if needed.
Black	Uses all pistes including black (expert) runs.

If no route can be found at your chosen difficulty, a red error message will appear telling you which leg failed and suggesting you try a higher difficulty.

3. The Route Panel

When a route is calculated, a panel slides up from the bottom of the screen showing the step-by-step directions.

Route Summary

The panel header shows key metrics for your route:

- Total distance (km) — including lift sections
- Skiing distance (km) — piste sections only
- Vertical drop (m) — total descent
- Duration (min) — estimated total time
- Maximum difficulty — shown as a coloured dot

Step-by-Step Directions

Each step shows an instruction such as "Take gondola Progetto Cabinovia to Station X" or "Ski down Alpino (red) to Station Y", along with the distance and duration for that segment.

Tap any step to highlight it on the map. The map will automatically zoom to show that segment. Tap again to deselect.

Dragging the Panel

The panel has three positions: collapsed (just the header), half-expanded (default), and fully expanded. Drag the handle bar at the top of the panel to resize it, or tap the header when collapsed to expand it.

Action Buttons

- Share — send the route via WhatsApp (see Section 11).
- Finished — log the route to your history (see Section 8).
- Clear — remove the current route and start over.

4. The Map

The map uses OpenTopoMap, a topographic map that shows terrain contours, which is ideal for understanding the mountain layout.

What You See on the Map

- Station markers — white circles with a blue outline. Tap to add as a stop.
- Pistes — coloured lines: blue, red, or dark grey/black matching their difficulty.
- Lifts — dashed grey lines.
- Route overlay — when a route is active, the steps are drawn as thicker coloured lines on top of the base map. Lifts appear as dashed lines, pistes as solid lines coloured by difficulty.
- Numbered stop markers — blue circles numbered 1, 2, 3… showing your route stops.
- Closed indicators — red "X" markers on any closed lifts or pistes included in your route.

Hovering Over Markers

Hover over (or long-press on mobile) any station marker to see a tooltip with the station name and elevation.

5. Switching Ski Areas

To switch between ski areas:

1. Tap the hamburger menu (three lines) in the top-right corner.
2. Tap "Ski Site".
3. You will see a list of available ski areas. Use the search box to filter, then tap the area you want.

When you switch areas, the map flies to the new area, all pistes and lifts update, and your current route is cleared. If you had previously cached a different area for offline use, you will be asked to confirm before switching.

6. GPS & Live Location

Tap the location button in the bottom-right corner of the map to enable GPS tracking. Your position appears as a blue dot with an accuracy circle around it.

When GPS is first activated, the map will fly to your current position. The button turns blue while GPS is active. Tap again to stop tracking.

Your browser will ask for location permission the first time. If permission is denied, an error message will appear near the button.

7. Daily Activity Tracking

Track your skiing day with the built-in activity recorder. Access it from the hamburger menu under "Daily Activity".

Recording

Tap "Start Recording" to begin tracking your movement via GPS. The app will automatically enable GPS if it is not already active. While recording:

- A red pulsing dot appears next to "Daily Activity" in the menu.
- Your total distance and maximum speed update in real time.
- Data is saved automatically every 10 GPS points and when you stop recording.

Tap "Stop Recording" when you are done. Your data is saved for the day and persists if you close and reopen the app.

Viewing Your Track on the Map

Check "Show track on map" to see your recorded track overlaid on the map. The track is colour-coded:

- Blue solid lines — skiing/downhill segments.
- Grey dashed lines — lift segments (detected by speed and altitude).

Replay

Tap "Play Replay" to watch an animated playback of your day. An orange marker moves along your track. Choose from 4 playback speeds: 1x, 2x, 5x, or 10x.

Reset

Tap "Reset" to clear the day's activity data. You will be asked to confirm before the data is deleted.

8. Run History & Season Summary

Logging a Run

After completing a route, tap "Finished" in the route panel to log it to your history. The run is saved with the date, route details, distance, vertical drop, and duration.

Viewing History

Open the hamburger menu and tap "History" to see all your logged runs, newest first. Each entry shows the start and end stations, distance, vertical drop, and time. Tap the X to delete an individual entry.

Season Summary

Open the hamburger menu and tap "Season Summary" to see aggregate statistics for all your logged runs:

- Total number of runs
- Total km skied
- Total vertical drop (m)
- Total time on the mountain
- Breakdown by difficulty (blue / red / black runs)
- Your top 3 most-repeated favourite routes

9. Weather Forecast

Open the hamburger menu and tap "Weather" to see the current conditions and a 7-day forecast for the selected ski area.

Current Conditions

Shows the current temperature, weather condition (with icon), wind speed and gusts, and the elevation of the weather station.

7-Day Forecast

Each day shows the high/low temperature, condition, precipitation or snowfall amount, and wind speed. Days are labelled "Today", "Tomorrow", or by weekday.

Tap "Refresh" to fetch the latest forecast. Weather data comes from the Open-Meteo service and is centred on the current ski area.

10. Piste & Lift Status

Open the hamburger menu and tap "P&L Status" to see which lifts and pistes are currently open or closed.

Each lift and piste is shown with a green "open" or red "closed" label. The header shows the count (e.g. "12/15 lifts, 30/35 pistes"). If everything is open, the panel shows "All pistes and lifts are open."

Closed lifts and pistes are automatically avoided by the route planner. If a closed segment must be used (no alternative exists), it will be shown with a red X marker on the map and a warning in the route panel.

Tap "Refresh" to fetch the latest status data.

11. Sharing a Route

With an active route, tap the "Share" button in the route panel. This opens WhatsApp with a pre-composed message containing:

- Start and end station names
- Number of stops (if multi-stop)
- Total distance, skiing distance, vertical drop, and duration
- A clickable link that recreates the exact route when opened

When the recipient opens the shared link, the app will automatically load the correct ski area, set the difficulty, and calculate the same route.

Note: Sharing requires an internet connection. If you are offline, nothing will happen.

12. Offline Use

The app is designed to work on the mountain where signal can be patchy. Once loaded, it works fully offline.

Automatic Caching

When you first load a ski area, the route data (stations, pistes, lifts) is cached locally. On subsequent visits, the app loads instantly from the cache even without internet.

Downloading Map Tiles

The map tiles (the background topographic map) are only cached as you browse. To ensure full map coverage offline:

4. Open the hamburger menu and tap "Offline Map".
5. Tap "Download Map Tiles". This pre-downloads all tiles for the current ski area at the relevant zoom levels.
6. Wait for the progress bar to complete.

It is recommended to do this on Wi-Fi before heading to the mountain.

Offline Indicator

When you lose internet connection, a brief amber banner appears at the top: "You are offline — using cached data". This auto-dismisses after 2 seconds.

What Works Offline

Feature	Offline?
Route planning & directions	Yes — fully functional
Map display	Yes — if tiles were cached or pre-downloaded
GPS & live location	Yes — GPS works without internet
Daily activity tracking	Yes — stored locally
Run history & season summary	Yes — stored locally
Weather & P&L status	No — requires internet to refresh

13. Tips & Troubleshooting

"No route found"

This means the router cannot find a path between your stops at the selected difficulty. The error message will tell you which leg failed. Try:

- Increasing the difficulty preference (e.g. from Blue to Red).
- Checking P&L Status — a key lift may be closed.
- Removing or reordering intermediate stops.

Map not showing pistes

If pistes are not visible after switching areas, try zooming in or out to trigger a map refresh. Ensure you have an internet connection for the initial load.

GPS not working

Ensure you have granted location permission to your browser. On iOS, this is under Settings > Safari > Location. On Android, check the site permissions in Chrome.

Installing as an App

On iOS: Open in Safari, tap the share button, then "Add to Home Screen".

On Android: Open in Chrome, tap the three-dot menu, then "Add to Home Screen" or "Install App".

Battery Saving

GPS tracking and daily activity recording use battery. If you do not need live tracking, tap the location button to turn off GPS when not in use.